

Fixing MSI / Intel Core Ultra Audio Issues on Kali Linux (SOF → HDA Fix)

Symptoms

- No sound from laptop speakers
 - OBS recordings had no microphone audio
 - HDMI audio devices only appeared
 - Internal speakers/microphone missing
 - PipeWire/PulseAudio defaulted to HDMI
 - `speaker-test` produced no sound
-

Diagnostic Commands Used

Check recording devices

```
arecord -l
```

Check playback devices

```
aplay -l
```

List available ALSA devices

```
arecord -L
```

Check PipeWire / PulseAudio status

```
pactl info
```

List audio sources

```
pactl list short sources
```

Test speaker output

```
speaker-test -c 2
```

Open ALSA mixer

```
alsamixer
```

Open PulseAudio/PipeWire control panel

```
pavucontrol
```

Check kernel audio logs

```
sudo dmesg | grep -iE "sof|snd|hda|audio"
```

Root Cause

Kali Linux loaded the Intel SOF audio driver stack (`sof-hda-dsp`) instead of the legacy Intel HDA driver.

This caused:

- HDMI-only audio routing
- missing analog speaker profiles
- microphone routing issues

Final Fix (Worked)

Force Linux to use legacy HDA audio instead of SOF DSP.

Create override

```
echo "options snd-intel-dspcfg dsp_driver=1" | sudo tee  
/etc/modprobe.d/alsa-base.conf
```

Rebuild initramfs

```
sudo update-initramfs -u
```

Reboot

```
sudo reboot
```

Result After Reboot

Audio devices correctly appeared as:

```
HDA Intel PCH  
ALC256 Analog  
analog-stereo
```

Internal speakers and microphone paths returned.

Verify After Reboot

```
pactl info
```

```
aplay -l
```

Rollback (If Needed)

Remove the override:

```
sudo rm /etc/modprobe.d/alsa-base.conf  
sudo update-initramfs -u  
sudo reboot
```

Notes

- System: MSI Modern 15 H AI C1MG
- CPU: Intel Core Ultra 5 125H
- OS: Kali Linux
- Audio Codec: Realtek ALC256
- Audio Stack: PipeWire + PulseAudio compatibility layer